เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of semester 2 **Year:** 2018

Subject: 394172 Mathematics 2 **Section:** 15-18

Date 26 Feb 2019 Time: 8:00-10:00

Name: ________________________ **ID:** __________________ **Class:** _______________

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Instructions:

1. The examination has 9 pages (including this page) and a total score of 90 points.
2. Write your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor permission.
6. You not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculations are NOT allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

No.	1.	2.	3.	4.	5.	6.	7.	8.	9
score									

Question 1 Let $p, q, r, s,$ and t be statements. Determine the following truth values: p is true and q is false. Find the truth value of the following statement:

1.1 $\left[(q \rightarrow s) \vee r\right] \vee \left[(p \leftrightarrow s) \wedge t\right]$. (5 points)

1.2 $\left[q \rightarrow (s \vee r)\right] \rightarrow \left[p \rightarrow (q \wedge \sim s)\right]$. (5 points)

Question 2 Show whether the following statements are logically equivalent or not.

2.1 $(p \to q) \vee (p \to r)$ and $p \to (q \vee r)$. (5 points)

2.2 $(p \wedge q) \to r$ and $(p \to \sim q) \wedge (p \to r)$. (5 points)

Question 3 3.1 Show whether the statement $(p \vee q) \wedge (\sim p \vee r) \rightarrow (q \vee r)$ is tautology or not. (5 points)

3.2 Given $U = R$, find the negation of the quantified statement $\forall x \forall y \left[x^2 + y^2 \geq 2xy\right]$ and determine whether the statement or its negation is true. (5 points)

Question 4 4.1 Given $U = R - \{0\}$. Find the truth value of the statement

$$\forall x \exists y\left[x^2 + \frac{1}{x^2} \geq y\right] \leftrightarrow \forall x \forall y\left[\sqrt{x^2 - 2xy + y^2} = x - y\right].$$ (5 points)

4.2 Given the argument is in the form

Hypothesis 1. $\sim s$

2. $p \rightarrow r$

3. $\sim r \rightarrow s$

Conclusion: $\sim p \vee (q \wedge s)$.

Show whether this argument is valid or not. (5 points)

Question 5 Convert the following numbers to a number with the base10. (10 points)

5.1 $11101.01_2 =$

5.2 $5A3E_{16} =$

5.3 $(3.456)_7 =$

Question 6. Convert the following numbers to a number with a given base. (10 points)

6.1 $4302_5 =$ (base 12)

6.2 $(A3.F)_{16} =$ (base 4)

Question 7. (10 points)

7.1 Convert the expression $(F21_{16} - 201_{8}) \times (320_{4} \div 24_{12})$ to a number with the base 14.

7.2 Solve the equation $(4114)_5 = (321)_x$ for x.

Question 8 8.1 Solve the absolute value inequality $1 < |5-3x| < 6$. (5 points)

8.2 Find all solutions of the absolute value inequality $\dfrac{|x|}{|x|-1} > 1$. (5 points)

Question 9 9.1 Find all solutions of $|3-5x|-|3x-5| < 1$. (5 points)

9.2 Find all solutions of $\left(x-|x-1|\right)\left(x^2-|x|\right) \geq 0$. (5 points)

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